MODDIV-WC V2: Unit-Anchored Hankel Rounding with Exact Fast Evaluation, Seeded Public-Key Compression, and a Conditional IND-CCA Route

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Revision and scope notice. This manuscript is A4 v1.2, a closure revision candidate for Cryptology ePrint 2026/2747. It corrects the perfect-correctness argument to use a signed modular discrepancy, adds exact FAST evaluation and a seeded COMPACT representation, makes the classical-ROM implicit-rejection composition explicit with a persistent fallback secret and a canonical reference encoding, and states a proof-route size barrier. The security conclusions are conditional on the precise binary-secret LWE source and on programmable classical random oracles where stated. No concrete 128-bit parameter set, standard-model theorem for SHAKE256, QROM theorem, formal verification, or deployment recommendation is claimed.

Abstract

We present a revised version of MODDIV-WC, a proof-oriented public-key encryption and key-encapsulation candidate based on a widened Hankel linear map followed by deterministic integer division. The construction uses the unit-anchored secret distribution $\{1\} \times \{Q, \ldots, 2Q-1\}^{d-1}$, for which the Hankel map is surjective for every admissible secret, together with an exact parity-mask embedding of a uniform binary secret.

This revision makes four points explicit. First, perfect correctness is proved with a signed modular discrepancy, avoiding an invalid bound on the difference of canonical representatives across wraparound. Second, an exact FAST identity evaluates the same rounded Hankel outputs without changing the cryptographic distribution. Third, a COMPACT representation replaces explicit uniformly random public Hankel inputs by a public seed in a dedicated programmable classical random oracle; its transmitted payload is $\kappa_A + tL(p-q)$ bits. Fourth, an implicit-rejection Fujisaki–Okamoto/HHK composition is stated with canonical encodings and conservative query accounting, yielding a single-user classical-ROM IND-CCA route conditional on the one-way bound of the base scheme.

The one-way analysis combines an ideal-key residual-hashing bound with a multiplicative order-two Renyi transfer from a specified binary-secret LWE source. A separate necessary-condition analysis shows that the current proof inequalities cannot certify a kilobyte-scale public key at target bias 2^{-128}: even a deliberately favorable relaxation requires at least 4,726,272 public-key bytes. This is a limitation of the present proof formula, not an attack or an unconditional lower bound. No concrete security level, real-hash theorem, QROM result, or security transfer to historical ModDiv schemes is asserted.

1 Introduction

Learning with errors (LWE) provides one of the best-understood worst-case-to-average-case foundations in modern cryptography [5, 6, 7]. Learning with rounding (LWR) removes explicit noise from the public function and replaces it by deterministic rounding [8]. The closeness of these two paradigms is deceptive: a naive proof that "noise rarely changes the rounded value"

can impose an unnecessarily strong ratio between the rounding cell width and the LWE noise. More refined reductions based on Renyi divergence avoid this loss and have been developed for LWR and structured middle-product variants [9, 10].

The public first version of MODDIV-WC appeared as Cryptology ePrint 2026/2747 on 3 November 2026 [3]. The present revision preserves its unit-anchored construction but corrects the modular correctness proof and reorganizes the FAST, COMPACT, conditional CCA, and parameter-size analyses. The earlier public version is therefore historical context rather than a premise for any proof below.

The present construction belongs to the ModDiv research lineage initiated by Bouftass. The original scalar proposal introduced the map

$$F(x) = ((ax) \bmod 2^p) \operatorname{Div}(2^q)$$

and used it as the basis of a public-key cryptosystem and key-exchange proposal [1]. A later multidimensional ModDiv proposal replaced the scalar product by a structured convolutional map and explicitly connected its security motivation to a learning-with-rounding problem [2]. The scalar proposal was subsequently analyzed by Zhang using a closest-vector formulation [4].

This work studies a new proof-oriented descendant of that line, called *MODDIV-WC*. Its public key contains pairs of uniformly random vectors and rounded outputs of a widened Hankel map. MODDIV-WC is *not* a retroactive security proof for the historical scalar or multidimensional ModDiv constructions: it changes the secret space and widens the Hankel representation specifically so that exact LWE embeddings, exact correctness, and a modular CCA proof become possible. The name therefore records conceptual lineage, while the security theorem applies only to the construction specified in this paper.

The main structural feature is a *unit anchor*: the first secret coordinate is fixed to one. For every admissible secret X, this makes

$$a \longmapsto \mathrm{Hank}_L(a, X)$$

a surjective homomorphism onto $\mathbb{Z}_P^L$. Consequently, a uniform input a produces an exactly uniform Hankel output. This simple fact is used twice: to establish exact distributional properties of the public key and to import a Renyi-divergence style LWE-to-rounding argument into the Hankel setting.

Relationship to prior ModDiv work. The modulo-then-integer-division primitive and the ModDiv naming convention originate in Bouftass's earlier work [1, 2]. The present construction is not a security proof for those historical schemes.

Relationship to MP-LWR/MP-CLWR and novelty scope

Middle-product LWE/LWR already provides the Toeplitz/Hankel convolutional viewpoint, deterministic rounding, full-rank structured-secret analysis, Renyi-divergence boundary arguments, and public-key encryption based on rounded middle products [10]. Randomized lifting of rounded values and reconciliation are likewise prior techniques. Accordingly, this paper does *not* claim novelty for Toeplitz/Hankel rounding, randomized lifting, Renyi boundary analysis, or reconciliation as general techniques.

The distinguishing mechanisms studied here are narrower. First, MODDIV-WC uses the secret distribution

$$\mathcal{S} = \{1\} \times \{Q, \ldots, 2Q-1\}^{d-1},$$

so that the fixed unit coordinate forces a triangular unit block and makes $a \mapsto \mathrm{Hank}_L(a, X)$ surjective for *every* admissible secret, rather than sampling and conditioning on a full-rank structured secret. Second, the binary-LWE embedding uses an exact parity mask $X = R + \iota(s)$

that maps a uniform binary secret to this box-secret distribution with zero statistical loss. To the best of our knowledge, we are not aware of a prior LWE/LWR construction combining these two mechanisms. This is a limited prior-art statement, not a claim of exhaustive novelty.

Technique	Prior status	Role in MODDIV-WC
Toeplitz/Hankel middle-product and rounding	Prior MP-LWE/MP-LWR	Reused structural setting
Randomized lifting and reconciliation	Prior LWR/MP-CLWR	Adapted to divisible power-of-two modules
Renyi boundary analysis	Prior BGM/MP-LWR	Adapted to the anchored secret law
Unit anchor $X_0 = 1$ for every-key surjectivity	No exact precedent found in our targeted search	Structural mechanism used here
Exact parity mask $R + \iota(s)$ into the box secret	No exact precedent found in our targeted search	Exact binary-LWE embedding
Implicit-rejection FO composition	FO/HHK is prior	Applied to the WC base construction

Contributions. The revision makes the following claims.

1. The BASE construction has perfect correctness under $t(Q-1) < P/8$, proved using a signed modular displacement rather than an ordinary difference of canonical representatives.

2. The unit anchor and sparse embedding yield an exact shared-secret binary-LWE simulation with a single private parity mask; the target box-secret distribution is obtained with zero statistical loss under independent uniform source bits.

3. FAST is an exact arithmetic identity for both the unrounded Hankel product and its quotient output. It introduces no cryptographic assumption or distributional loss.

4. COMPACT represents the explicit uniformly random public Hankel inputs by a short public seed in a dedicated programmable classical random oracle. In a fresh seed context the coupling is exact; prequeries cost at most ε_A.

5. The ideal-key residual-hashing bound and the anchored Renyi-transfer bound give the conditional one-way estimate

$$p_{\mathrm{OW}} \le \sqrt{\overline{D}_R \left(\mathrm{Adv}^{\mathsf{binLWE}} + 2^{-\kappa} + \varepsilon_{\mathrm{enc}} \right)}.$$

6. With canonical encodings and independent oracle domains, the compact FAST KEM satisfies the conditional single-user classical-ROM bound

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{C,\mathrm{FAST}} \le q_H 2^{-\kappa} + N \left[\sqrt{\overline{D}_R \left(\mathrm{Adv}^{\mathsf{binLWE}} + 2^{-\kappa} + \varepsilon_{\mathrm{enc}} \right)} + \varepsilon_A \right].$$

7. The exact residual threshold improves the asymptotic compact-key growth of the illustrative family, but the present numerical proof formula still imposes a large payload. At target bias 2^{-128} the relaxed necessary payload is 4,726,272 bytes.

Scope. Version 2 does not claim competitive concrete parameters. The FAST identity is exact; no portable speed factor is claimed here. COMPACT is exact in the specified programmable classical-ROM experiment; publishing a short seed for a fixed deterministic XOF does not by itself reproduce that proof, and SHAKE256 is treated only as a heuristic instantiation. The CCA theorem is single-user and classical-ROM only. The parameter lower bound is a lower

bound on what the *current inequalities can certify*; it is not an impossibility theorem for every future proof or every ModDiv-derived construction. No QROM, side-channel, multi-user, or concrete 128-bit claim is made.

2 Preliminaries and Notation

2.1 Modular representatives and rounding

Let $P = 2^p$ and $Q = 2^q$ with $Q \mid P$. We write

$$H = P/Q.$$

For $z \in \mathbb{Z}_P$, let $[z]_P \in \{0, \ldots, P-1\}$ denote its canonical representative and define the cell-division map

$$F_Q(z) = \left\lfloor \frac{[z]_P}{Q} \right\rfloor \in \{0, \ldots, H-1\}.$$

The map is applied coordinatewise to vectors.

2.2 Widened Hankel map

For

$$a = (a_0, \ldots, a_{m-1}) \in \mathbb{Z}_P^m, \qquad X = (X_0, \ldots, X_{d-1}) \in \mathbb{Z}_P^d,$$

define

$$\mathrm{Hank}_L(a, X)_i = \sum_{j=0}^{d-1} a_{i+j} X_j \pmod{P}, \qquad 0 \le i < L.$$

The dimensions used throughout are

$$d = (h-1)L + 2, \qquad m = hL + 1.$$

2.3 Multiplicative Renyi divergence

For distributions P_0, P_1 on a finite set with P_0 absolutely continuous with respect to P_1, define the multiplicative order-2 Renyi divergence by

$$\mathrm{RD}_2(P_0 \| P_1) = \sum_x \frac{P_0(x)^2}{P_1(x)}.$$

This convention is the one used in the LWR probability-preservation arguments relevant here [9, 10]. Two facts will be used repeatedly:

$$\Pr_{P_0}[E]^2 \le \mathrm{RD}_2(P_0 \| P_1) \Pr_{P_1}[E], \tag{1}$$

$$\mathrm{RD}_2(P_0^{\otimes r} \| P_1^{\otimes r}) = \mathrm{RD}_2(P_0 \| P_1)^r. \tag{2}$$

Renyi divergence also obeys data processing under a common randomized channel.

2.4 Security notions

We use the following conventions throughout. The quantity $\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}$ is a *success probability*: the probability of recovering the uniformly sampled hidden message in the one-way chosen-plaintext experiment. For binary-secret LWE, $\mathrm{Adv}^{\mathsf{binLWE}}$ is the absolute acceptance-probability gap between the noisy source and its stated ideal source. For the KEM,

$$\mathrm{Adv}^{\mathsf{IND\text{-}CCA}} := \left| \Pr[b' = b] - \frac{1}{2} \right|$$

is the IND-CCA *success bias*. Thus the difference between the two conditioned output branches is twice this bias. All bounds below use these conventions explicitly; no factor-of-two conversion is left implicit.

3 The MODDIV-WC Construction

3.1 Parameters and secret space

All arithmetic parameters in this paper are integers. Choose

$$q \geq 1, \qquad p \geq q+3, \qquad P = 2^p, \qquad Q = 2^q, \qquad H = P/Q,$$

with $h, L, t \geq 1$ and $1 \leq \kappa \leq L$. The random-oracle output lengths κ_A, κ_K are positive integers. Let h be the binary-LWE secret dimension, L the number of Hankel outputs per public block, t the number of public blocks, and κ the payload length. Every theorem below uses this admissible domain unless it explicitly states a smaller one. Define

$$d = (h-1)L + 2, \qquad m = hL + 1.$$

The secret space is

$$\mathcal{S} = \{1\} \times \{Q, \ldots, 2Q-1\}^{d-1}.$$

Thus every secret satisfies

$$X_0 = 1.$$

3.2 Key generation

Sample

$$X_0 = 1, \qquad X_i \leftarrow U\{Q, \ldots, 2Q-1\} \quad (1 \leq i < d).$$

For each $j \in \{0, \ldots, t-1\}$ sample

$$a^{(j)} \leftarrow U(\mathbb{Z}_P^m),$$

compute

$$Y^{(j)} = \mathrm{Hank}_L(a^{(j)}, X),$$

and publish

$$U^{(j)} = F_Q(Y^{(j)}).$$

The public and secret keys are

$$pk = \left((a^{(j)}, U^{(j)})\right)_{j=0}^{t-1}, \qquad sk = X.$$

3.3 Encryption

For a message $M \in \{0,1\}^\kappa$, sample

$$r = (r_0, \ldots, r_{t-1}) \leftarrow U(\{0,1\}^t)$$

and, for each j, i,

$$\rho_i^{(j)} \leftarrow U\{0, \ldots, Q-1\}.$$

Lift each public quotient by

$$u_i^{(j)} = QU_i^{(j)} + \rho_i^{(j)}.$$

Then compute

$$c_1 = \left[\sum_{j=0}^{t-1} r_j a^{(j)}\right]_P \in \mathbb{Z}_P^m,$$

$$v = \left[\sum_{j=0}^{t-1} r_j u^{(j)} \right]_P \in \mathbb{Z}_P^L.$$

Define the reconciliation maps

$$K_P(z) = \left\lfloor \frac{[z + P/4]_P}{P/2} \right\rfloor \in \{0, 1\},$$

$$C_P(z) = \left\lfloor \frac{[z]_P}{P/4} \right\rfloor \bmod 2,$$

applied coordinatewise. Let

$$z = K_P(v), \qquad c_2 = C_P(v), \qquad c_3 = M \oplus z_{0..\kappa-1}.$$

The ciphertext is

$$c = (c_1, c_2, c_3).$$

3.4 Decryption and reconciliation

Given $c = (c_1, c_2, c_3)$, compute

$$w = \mathrm{Hank}_L(c_1, X).$$

For $a_8 = P/8$, define

$$\mathrm{Rec}_P(y, b) = 0 \iff [y + (2b + 1)a_8]_P \leq P/2 - 2.$$

Apply Rec_P coordinatewise to obtain z', and return

$$M' = c_3 \oplus z'_{0..\kappa-1}.$$

The reconciliation rule is designed so that if

$$y \equiv z + \Delta \pmod{P}, \qquad |\Delta| < P/8,$$

then

$$\mathrm{Rec}_P(y, C_P(z)) = K_P(z).$$

Theorem 3.1 (Perfect correctness)**.** *If*

$$t(Q - 1) < P/8,$$

then MODDIV-WC decrypts every honestly generated ciphertext correctly.

Proof. Write

$$Y_i^{(j)} = QU_i^{(j)} + \tau_i^{(j)}, \qquad 0 \leq \tau_i^{(j)} < Q.$$

Because $u_i^{(j)} = QU_i^{(j)} + \rho_i^{(j)}$,

$$u_i^{(j)} - Y_i^{(j)} = \rho_i^{(j)} - \tau_i^{(j)}.$$

Linearity of the Hankel map gives

$$w_i \equiv \sum_j r_j Y_i^{(j)} \pmod{P}.$$

Define the signed integer displacement

$$\delta_i = \sum_j r_j (\rho_i^{(j)} - \tau_i^{(j)}).$$

Then

$$v_i \equiv w_i + \delta_i \pmod{P}, \qquad |\delta_i| \leq \sum_j r_j (Q - 1) \leq t(Q - 1) < P/8.$$

The reconciliation property applies with $z = v_i$, $y = w_i$ and displacement $-\delta_i$, so encryption and decryption recover the same key bits. In particular, no bound on $|[v_i]_P - [w_i]_P|$ is needed or valid across modular wraparound. □

Why the signed formulation is necessary. Take $P = 32$, $Q = 2$, $t = 2$, $X = (1, 2)$, public blocks $a^{(0)} = (1, 0)$ and $a^{(1)} = (31, 0)$, selectors $r_0 = r_1 = 1$, and lift remainders equal to zero. Then $v = 30$ and $w = 0$, while the signed discrepancy is $\delta = -2$. Thus the canonical difference has magnitude 30, but the modular displacement has magnitude $2 < P/8$ and reconciliation succeeds. This finite witness rules out the stronger canonical-difference assertion without affecting perfect correctness.

4 Exact FAST Evaluation

Version 2 separates the arithmetic implementation from the cryptographic distribution. Write every non-anchor secret coordinate as

$$X_j = Q + s_j, \qquad 0 \le s_j < Q, \qquad 1 \le j < d.$$

For each output coordinate define the integer sums

$$A_i = \sum_{j=1}^{d-1} a_{i+j}, \qquad S_i = a_i + \sum_{j=1}^{d-1} a_{i+j} s_j.$$

Theorem 4.1 (Exact FAST identity)**.** *For $P = QH$ and every admissible a, X,*

$$\boxed{F_Q(\mathrm{Hank}_L(a, X)_i) = \left[A_i + \left\lfloor \frac{S_i}{Q} \right\rfloor \right]_H .}$$

Consequently the FAST evaluator is functionally identical to the baseline rounded Hankel evaluator for every admissible input.

Proof. Let

$$T_i = \sum_{j=0}^{d-1} a_{i+j} X_j = QA_i + S_i$$

as an integer. Divide the complete low-part sum as

$$S_i = Qb_i + r_i, \qquad 0 \le r_i < Q,$$

and write

$$A_i + b_i = Hk_i + c_i, \qquad 0 \le c_i < H.$$

Then

$$T_i = Pk_i + Qc_i + r_i.$$

Since $0 \le Qc_i + r_i < P$, the canonical representative of T_i modulo P is exactly $Qc_i + r_i$, and division by Q returns c_i. This is the claimed expression. □

The theorem is a carry statement: the low q bits may be discarded only *after* their joint carry has been incorporated through $\lfloor S_i/Q \rfloor$. In particular, termwise truncation is invalid in general. One may safely keep A_i modulo H and S_i modulo P, because the discarded multiples become multiples of P or H at the final output.

The scalar-width gain should not be overstated. The baseline non-anchor multipliers X_j already have at most $(q + 1)$ bits, whereas FAST uses $s_j < Q$, so the scalar decomposition reduces that operand by at most one bit. The theorem is therefore an exact functional result, not a general performance claim; implementation speed depends on arithmetic widths, vectorization, memory layout, and platform.

5 Exact Binary-LWE Embedding

Let

$$A \in \mathbb{Z}_P^{L\times h}.$$

Construct $a \in \mathbb{Z}_P^m$ by choosing a_0 uniformly and setting

$$\boxed{a_{1+vL+i} = A_{i,v}} \qquad (0 \le v < h,\ 0 \le i < L).$$

For $s \in \{0,1\}^h$, define the sparse vector $\iota(s) \in \mathbb{Z}_P^d$ by

$$\iota(s)_{1+vL} = s_v$$

and zero elsewhere.

Lemma 5.1 (Exact Hankel embedding)**.** *For every A and s defined above,*

$$\mathrm{Hank}_L(a, \iota(s)) = As.$$

Proof. For output coordinate i,

$$\mathrm{Hank}_L(a, \iota(s))_i = \sum_{v=0}^{h-1} a_{i+1+vL} s_v = \sum_{v=0}^{h-1} A_{i,v} s_v.$$

□

The binary secret is converted to the exact MODDIV-WC secret distribution by a parity mask. Set $R_0 = 1$. At the active coordinates sample

$$R_{1+vL} \leftarrow U\{Q, Q+2, \ldots, 2Q-2\},$$

and at all other tail coordinates sample uniformly from $\{Q, \ldots, 2Q-1\}$. Define

$$X = R + \iota(s).$$

Since Q is even, every value of $\{Q, \ldots, 2Q-1\}$ has a unique representation as an even base point plus one binary parity bit. Hence X has exactly the key-generation law.

Now suppose a binary-LWE block is

$$b = As + E \pmod P.$$

The simulator forms

$$y = b + \mathrm{Hank}_L(a, R) \pmod P,$$

which by Lemma 5.1 satisfies

$$\boxed{y = \mathrm{Hank}_L(a, X) + E \pmod P.}$$

Thus the real binary-LWE branch yields a noisy MODDIV-WC public block with the exact target secret law, while a uniform binary-LWE branch yields a uniform y independent of a.

6 The Unit Anchor and Uniform Hankel Outputs

Lemma 6.1 (Unit-anchor surjectivity)**.** *For every admissible secret $X \in \mathcal{S}$, the map*

$$T_X : \mathbb{Z}_P^m \to \mathbb{Z}_P^L, \qquad T_X(a) = \mathrm{Hank}_L(a, X),$$

is surjective.

Proof. Fix arbitrary tail variables $a_L, \ldots, a_{m-1}$ and a target $y \in \mathbb{Z}_P^L$. Because $X_0 = 1$, the equations

$$y_i = a_i + \sum_{j=1}^{d-1} a_{i+j} X_j$$

can be solved successively for $a_{L-1}, a_{L-2}, \ldots, a_0$. Thus every target has a preimage. □

Corollary 6.2 (Exact uniformity)**.** *For fixed admissible X and uniform $a \leftarrow U(\mathbb{Z}_P^m)$,*

$$\mathrm{Hank}_L(a, X) \leftarrow U(\mathbb{Z}_P^L).$$

In particular, the L output coordinates are jointly independent and uniform.

Proof. T_X is a surjective homomorphism of finite abelian groups, so every fiber has the same cardinality. □

7 Seeded Public Randomness and Compact Public Keys

The explicit BASE public key contains t uniformly random vectors $a^{(j)} \in \mathbb{Z}_P^m$, which dominate its serialized size. Version 2 introduces a dedicated classical programmable random oracle $\mathcal{A}$, domain-separated from the FO oracles below. For a public seed $\sigma_A \in \{0,1\}^{\kappa_A}$ and block index j, let $D(\theta, \sigma_A, j)$ be an injective encoding of the public parameter context, seed, and index. Read the first mp oracle bits at that input as m consecutive p-bit fields and denote the resulting vector by

$$a^{(j)} = \mathrm{Expand}_{\mathcal{A}}(\theta, \sigma_A, j).$$

Because $P = 2^p$, no rejection sampling or modular reduction is needed.

Theorem 7.1 (Exact seeded public randomness in the ROM)**.** *Conditioned on a fresh seed context, the expanded family satisfies*

$$(a^{(0)}, \ldots, a^{(t-1)}) \sim U((\mathbb{Z}_P^m)^t)$$

exactly. Hence, after computing the same deterministic quotients $U^{(j)}$, the developed view of the compact key has exactly the BASE key-generation distribution.

Proof. For a fixed block, mp independent uniform bits are in bijection with m independent elements of $\mathbb{Z}_{2^p}$. Distinct block encodings give distinct random-oracle inputs, so the block prefixes are independent. Their joint point probability is therefore P^{-tm}, exactly the uniform mass on $(\mathbb{Z}_P^m)^t$. □

The transmitted public key becomes

$$\boxed{pk_C = (\sigma_A, U^{(0)}, \ldots, U^{(t-1)})}$$

with tightly packed size

$$\boxed{|pk_C| = \kappa_A + tL(p-q) \quad \text{bits.}}$$

The explicit BASE representation has size

$$|pk_{\mathrm{BASE}}| = t(mp + L(p-q)).$$

For the asymptotic family of Version 1 this removes the tmp term and changes $\Theta(\lambda^7 \log \lambda)$ to $\Theta(\lambda^5 \log \lambda)$ before the further parameter diet of Section 13.

Programmability. In the main single-target game the simulator first receives or forms the jointly uniform BASE blocks, then samples σ_A, programs the fresh prefixes $\mathcal{A}(D(\theta, \sigma_A, j))$ to those blocks, and only then publishes the compact key. Because the programmed values have exactly the honest marginal law, this coupling has

$$\varepsilon_A = 0.$$

If the interface permits oracle queries before the seed is sampled, and d_A^{pre} distinct seeds in the same context have already exposed at least one target block, then

$$\Pr[\mathrm{Bad}_A \mid \mathrm{view}] = \frac{d_A^{\mathrm{pre}}}{2^{\kappa_A}} \le \min\left(1, \frac{q_A^{\mathrm{pre}}}{2^{\kappa_A}}\right).$$

Queries after publication are not bad events: the programmed answer is already fixed and is returned consistently.

Concrete XOF caveat. A short public seed does not information-theoretically encode *tmp* independent bits for a fixed deterministic function. In particular, the usual hidden-seed PRG statement $F(U_{\kappa_A}) \approx_c U_N$ cannot justify the joint view $(\sigma_A, F(\sigma_A))$, because a distinguisher can recompute $F(\sigma_A)$. SHAKE256 is therefore a functional and reproducible instantiation of the expander, but the exact proof above is a programmable-ROM statement rather than a standard-model proof for SHAKE256.

Measured size effect. On the representation-only profile $(p, q, m, L, t) = (256, 128, 513, 32, 64)$ with a 256-bit seed, the tightly packed public payload fell from 1,083,392 to 32,800 bytes, a factor of about 33.03. This profile was designed to measure the representation and is not a concrete-security parameter set.

8 Ideal Public Keys and Residual Hashing

In the uniform binary-LWE branch, each translated signal $y^{(j)}$ is uniform in $\mathbb{Z}_P^L$. Since $Q \mid P$, the public quotient

$$F_Q(y^{(j)})$$

is exactly uniform in $\mathbb{Z}_H^L$. We call this the *ideal public-key world*.

In that world, after drawing fresh $\rho^{(j)}$, the lifted vectors

$$u^{(j)} = QU^{(j)} + \rho^{(j)}$$

are uniform in $\mathbb{Z}_P^L$. The encryption map

$$r \longmapsto \sum_j r_j u^{(j)}$$

is 2-universal because every nonzero difference $r - r'$ contains a coefficient ± 1. The leakage

$$c_1 = \sum_j r_j a^{(j)}$$

reveals at most $m \log_2 P$ bits in the collision-probability sense. A standard strong leftover-hash argument therefore gives the bound used below.

Lemma 8.1 (Residual hashing with the full leakage)**.** *Let $r \leftarrow U(\{0,1\}^t)$, let*

$$c_1 = \sum_j r_j a^{(j)}, \qquad v = \sum_j r_j u^{(j)} \pmod P,$$

and let $V_0 \leftarrow U(\mathbb{Z}_P^L)$ be independent. In the ideal public-key world,

$$\mathrm{TV}\big((a,u,c_1,v),(a,u,c_1,V_0)\big) \le \boxed{\varepsilon_{\mathrm{enc}} := \frac{1}{2}\sqrt{\frac{P^{m+L}}{2^t}}}.$$

Here $a = (a^{(0)},\dots,a^{(t-1)})$ and $u = (u^{(0)},\dots,u^{(t-1)})$; revealing the complete lift u is stronger than revealing only the public quotients $F_Q(u)$.

Proof. Fix the public ribbons a. For each value c in the image of $r \mapsto c_1$, let S_c be its nonempty selector fiber and write $s_c = |S_c|$. Conditional on $c_1 = c$, the selector is uniform on S_c, while the complete lift u is an independent uniform seed. The two-universal subsum family therefore gives the strong leftover bound

$$\mathbb{E}_u\ \mathrm{TV}\big(\mathcal{L}(h_u(r) \mid c_1 = c, u), U(\mathbb{Z}_P^L)\big) \le \frac{1}{2}\sqrt{\frac{P^L}{s_c}}.$$

Weighting by $s_c/2^t$ and summing over fibers gives

$$\frac{\sqrt{P^L}}{2^{t+1}} \sum_c \sqrt{s_c} \le \frac{1}{2}\sqrt{\frac{P^{m+L}}{2^t}},$$

because there are at most P^m possible values of c_1 and $\sum_c s_c = 2^t$. Averaging over a proves the joint statement, with both u and c_1 retained. □

Now add an independent uniform message $M \leftarrow U(\{0,1\}^\kappa)$ and apply the public ciphertext channel. In the comparison world, for uniform V_0 the pair $(K_P(V_0), C_P(V_0))$ is coordinatewise uniform; hence the first κ key bits one-time-pad M even given the helper bits. Therefore any one-way adversary seeing the public key and challenge ciphertext has ideal-world success at most $2^{-\kappa}$. Data processing and the lemma yield

$$\boxed{p_{\mathrm{ideal}} \le 2^{-\kappa} + \varepsilon_{\mathrm{enc}}.}$$

This is a statement for the OW experiment with hidden uniform M; it does not claim that the mask remains hidden after conditioning on a fixed known message and its c_3.

9 Renyi Transfer for the Unit-Anchored Distribution

The earlier statistical proof compared clean and noisy public keys by a union bound, paying approximately $tL\,\mathbb{E}|E|/Q$. That bound can force Q to dominate the noise by a negligible factor. The following lemma replaces statistical closeness by multiplicative Renyi divergence, following the probability-preservation strategy used for LWR and middle-product LWR [9, 10].

Fix an admissible X. Define one clean public block

$$\mathcal{C}_X = (a, F_Q(\mathrm{Hank}_L(a, X)))$$

and one noisy public block

$$\mathcal{N}_X = (a, F_Q(\mathrm{Hank}_L(a, X) + E))\,.$$

Here $a \leftarrow U(\mathbb{Z}_P^m)$. The error vector is sampled independently of the public ribbon a, the secret X, and the source variables that determine X; its coordinates are mutually independent. More generally, one may condition on a latent variable Λ that is independent of (a, X) and of those source variables, provided that, for every value of Λ with positive probability, the coordinates of E are conditionally independent and the symmetry and tail bounds below hold uniformly. In particular, the error law may not adapt to the realized residue $\mathrm{Hank}_L(a, X)$.

For $0 \le r < P$, write $u = r \bmod Q$. Under $2B < Q$, call r *B-dangerous* exactly when

$$u \in \{0, \dots, B-1\} \cup \{Q-B, \dots, Q-1\}.$$

Thus each complete quotient cell has exactly $2B$ dangerous residues.

Theorem 9.1 (Masked-Hankel Renyi transfer)**.** *Assume the joint error law just specified. For every coordinate, conditionally on every allowed Λ, suppose the law is symmetric about zero and satisfies*

$$\Pr(|E_i| \le B \mid \Lambda) \ge \delta_E,$$

with common deterministic bounds B, δ_E satisfying $2B < Q$. Then

$$\boxed{\mathrm{RD}_2(\mathcal{C}_X \| \mathcal{N}_X) \le \left(\frac{1 + 2B/Q}{\delta_E}\right)^L.}$$

For t independent public blocks sharing the same fixed secret X,

$$\boxed{\overline{D}_R \le \left(\frac{1 + 2B/Q}{\delta_E}\right)^{tL}.}$$

The same bound holds after sampling X from the MODDIV-WC key-generation distribution.

Proof. By Corollary 6.2,

$$Z = \mathrm{Hank}_L(a, X)$$

is uniform in $\mathbb{Z}_P^L$. By the definition above, every quotient cell contains exactly $2B$ dangerous residues, so the fraction of dangerous residues is

$$\theta \le \frac{2BH}{P} = \frac{2B}{Q}.$$

For a fixed z, let $r(z)$ denote the number of dangerous coordinates. If a coordinate is not dangerous, the rounded value is unchanged whenever $|E_i| \le B$, an event of probability at least δ_E. If a coordinate is dangerous, exactly one of the two half intervals $[-B, 0]$ or $[0, B]$ points toward the interior of the current cell. Symmetry implies that this favorable half interval has probability at least $\delta_E/2$. Independence of the error coordinates therefore gives

$$\Pr[F_Q(z + E) = F_Q(z)] \ge \delta_E^L 2^{-r(z)}.$$

Because the clean distribution is deterministic given a, the multiplicative order-2 Renyi divergence can be written as

$$\mathrm{RD}_2(\mathcal{C}_X \| \mathcal{N}_X) = \mathbb{E}_a \left[\frac{1}{\Pr_E(F_Q(Z + E) = F_Q(Z) \mid Z)} \right].$$

Hence

$$\mathrm{RD}_2(\mathcal{C}_X \| \mathcal{N}_X) \le \delta_E^{-L}\, \mathbb{E}_Z[2^{r(Z)}].$$

The L coordinates of Z are independent and uniform, so

$$\mathbb{E}_Z[2^{r(Z)}] \le (1+\theta)^L \le \left(1+\frac{2B}{Q}\right)^L.$$

This proves the one-block bound. The t-block bound follows from multiplicativity conditionally on X because all tL error coordinates are independent under the stated joint law. Revealing X (and, when present, Λ) in both experiments and then forgetting those analytic variables by data processing preserves the same bound for the actual mixed distributions. Since the bound is uniform after conditioning on Λ, averaging the conditional laws introduces no additional factor. A public-dependent noise rule is outside the theorem: marginal symmetry alone is insufficient. □

Remark 9.2 (Useful asymptotic regime)**.** *If*

$$\frac{2BtL}{Q} = o(1) \qquad \textit{and} \qquad tL(1-\delta_E) = o(1),$$

then

$$\log \overline{D}_R \le tL\,(\log(1+2B/Q) - \log \delta_E) = o(1),$$

so

$$\overline{D}_R = 1 + o(1).$$

This is the analogue, for the present anchored Hankel map, of the bounded Renyi-loss regime used in small-modulus LWR reductions [9, 10].

10 From Binary-LWE to OW-CPA Security

Let p_{clean} be the success probability of an OW-CPA adversary against an honestly generated MODDIV-WC key, and let p_{noisy} be its success probability when the public key is generated from the noisy binary-LWE embedding.

The public key, the challenge encryption, and the adversary's computation form a common randomized channel applied to the clean and noisy key distributions. By data processing and Eq. (1), Theorem 9.1 gives

$$\boxed{p_{\text{clean}} \le \sqrt{\overline{D}_R p_{\text{noisy}}}.}$$

A binary-LWE distinguisher can generate the OW challenge itself and output one precisely when the OW adversary recovers the sampled message. Therefore

$$p_{\text{noisy}} \le \mathrm{Adv}^{\mathsf{binLWE}} + p_{\text{ideal}}.$$

Combining with Lemma 8.1 yields the following.

Theorem 10.1 (OW-CPA reduction)**.** *For MODDIV-WC,*

$$\boxed{\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathrm{WC}} \le \sqrt{\overline{D}_R\left(\mathrm{Adv}^{\mathsf{binLWE}} + 2^{-\kappa} + \varepsilon_{\text{enc}}\right)}.}$$

This theorem is the key reason that the proof does not need a generic decisional Hankel-LWR theorem. Renyi divergence preserves a one-way success probability directly, and the implicit-rejection FO transform used below has a published OW-CPA route.

11 Implicit-Rejection Fujisaki–Okamoto Wrapper and Compact CCA Preservation

Version 2 uses three independent classical random-oracle domains: the public expansion oracle $\mathcal{A}$ of Section 7, the FO coin oracle

$$G : \mathcal{M} \to \{0,1\}^{t(1+Lq)},$$

and the key oracle

$$H : \mathcal{M} \times \mathcal{C}_{\mathrm{wire}} \to \{0,1\}^{\kappa_K}.$$

Their encodings are injectively domain-separated. The message space is $\mathcal{M} = \{0,1\}^{\kappa}$.

KEM key generation and persistent rejection state. The KEM key generator first runs the compact PKE key generation to obtain (pk_C, X), then samples one independent fallback string

$$s \leftarrow U(\mathcal{M})$$

once per key. It returns pk_C and stores $sk_{\mathrm{KEM}} = (X, s, pk_C)$; an API that supplies pk_C separately may equivalently store only (X, s). The same s is reused deterministically for every rejection under that secret key. This is the persistent random seed of the implicit-rejection interface of HHK Figure 12, not fresh decapsulation randomness.

Encapsulation samples $\mu \leftarrow U(\mathcal{M})$, computes

$$c = \mathrm{Enc}(pk_C, \mu; G(\mu)), \qquad c_{\mathrm{wire}} = \mathrm{EncodeCT}(c), \qquad K = H(\mu, c_{\mathrm{wire}}),$$

and returns (c_{wire}, K). Decapsulation receives the raw byte string c_{wire}. If strict parsing fails, it returns $H(s, c_{\mathrm{wire}})$. Otherwise it decrypts the parsed ciphertext to a candidate μ', re-encrypts with the deterministic tape $G(\mu')$, and compares the *complete canonical bytes.* On exact equality it returns $H(\mu', c_{\mathrm{wire}})$; on inequality it returns $H(s, c_{\mathrm{wire}})$. No validity bit is returned. Repeated rejection of the same byte string under one secret key therefore returns the same random-oracle value.

Reference canonical encoding. For a fixed parameter profile, all fields have fixed widths and are packed little-endian at the bit level. A compact public key consists of the κ_A seed bits followed by the tL quotient fields, each of width $p - q$. A ciphertext consists of the m coordinates of c_1, each of width p, followed by the L helper bits c_2 and the κ payload bits c_3. The total byte length is fixed; every unused high padding bit in the last byte must be zero. Decoding rejects a wrong length, nonzero padding, or a field outside its declared range before any semantic processing. Consequently every accepted object has exactly one byte encoding. The IND-CCA decapsulation oracle refuses the exact challenge byte string; strict parsing prevents an alternative padding alias from representing the same accepted ciphertext. Public-key imports use the analogous strict compact-key parser and are rejected rather than repaired when malformed.

FAST changes no value by Theorem 4.1. For a fixed expansion oracle, expanding pk_C therefore yields exactly the same PKE and decapsulation behavior as the explicit representation, for every canonical input and every malformed byte string handled by the wrapper.

For a perfectly correct base PKE, the corrected HHK modular analysis supplies the classical-ROM OW-to-CCA interface used here. The COMPACT representation adds only the coupling of Section 7. We use the principal HHK game and do not claim an additional pre-challenge decapsulation extension. Let q_G be an upper bound on all G-oracle evaluations made explicitly by the reduction/adversary in the deterministic-encryption stage. Let q_H bound the adversary's H queries; Theorem 3.4 of HHK turns these into at most q_H plaintext-checking calls, each of which may induce a G evaluation in the Theorem 3.1 simulation. Repeated queries may be

cached, but the conservative budget counts them unless a distinct-query bound is stated. The corrected Theorem 3.1 coefficient is therefore

$$N = q_G + q_H + 1,$$

where the final +1 covers the output/recovery step. Decapsulation-query count affects running time and table operations; it is not silently omitted from the resource budget.

Budget	Meaning in this composition
q_G	explicit/induced evaluations of the FO coin oracle accounted in the T-stage simulation
q_H	adversarial key-oracle queries; also an upper bound on plaintext-checking calls produced by the U-stage reduction
q_D	decapsulation queries; charged to running time and simulation work
q_A^{pre}	expansion-oracle seed contexts exposed before the compact seed is sampled; contributes only through ε_A

Theorem 11.1 (Compact FAST CCA preservation)**.** *Under the BASE OW-CPA premise established above, in the single-user classical programmable-ROM experiment,*

$$\boxed{\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{C,\mathrm{FAST}} \le q_H 2^{-\kappa} + N\left(\mathrm{Adv}^{\mathsf{OW\text{-}CPA}}_{\mathrm{BASE}} + \varepsilon_A\right).}$$

In the principal fresh-context experiment, $\varepsilon_A = \varepsilon_{\mathrm{FAST}} = 0$. If expansion-oracle prequeries are allowed before the seed is chosen, the separately defined compact-key coupling uses $\varepsilon_A \le \min(1, q_A^{\mathrm{pre}}/2^{\kappa_A})$; this is not a decapsulation-phase loss.

Combining the theorem with the Renyi-transfer OW bound yields

$$\boxed{\mathrm{Adv}^{\mathsf{IND\text{-}CCA}}_{C,\mathrm{FAST}} \le q_H 2^{-\kappa} + N\left[\sqrt{\overline{D}_R\left(\mathrm{Adv}^{\mathsf{binLWE}} + 2^{-\kappa} + \varepsilon_{\mathrm{enc}}\right)} + \varepsilon_A\right].}$$

No new cryptographic assumption is introduced by FAST. The expansion oracle $\mathcal{A}$ is an explicit idealization. The result is not a standard-model proof for SHAKE256, is not a multi-user theorem, and makes no QROM claim.

12 External Binary-LWE Obligation and Asymptotic Compatibility

The security theorem above is conditional on hardness of the precise binary-secret LWE source at the reduction's resource budget. Public LWE literature provides worst-case foundations and binary-secret/modulus transformations [5, 6, 7], but this paper does not re-prove or certify their complete composition for any finite MODDIV-WC profile. The source dimension n_0, binary-secret dimension h, source modulus q_0, and final WC modulus P must remain distinct.

For example, a binary-secret transformation of Brakerski et al. requires conditions of the form

$$q_0 \ge 2^{n_0/2}, \qquad h \ge (n_0+1)\log_2 q_0 + 2\log_2(1/\delta_B),$$

together with its own sample, noise, and statistical-loss hypotheses. Replacing the source width $\log_2 q_0$ by the smaller final WC width p before the relevant modulus transformation would change the premise.

12.1 A conditional asymptotic family

Take $n_0 = \lambda$, $q_0 = 2^{\lceil \lambda/2 \rceil}$ and $\alpha_0 = \lambda^{-12}$. Choose exponentially small auxiliary losses, for example $\delta_B = 2^{-4\lambda}$, so that one may take

$$h = (\lambda + 1)\lceil \lambda/2 \rceil + 8\lambda = \Theta(\lambda^2).$$

For the MODDIV-WC layer choose

$$P = 2^{\lceil 16 \log_2 \lambda \rceil}, \qquad Q = 2^{\lceil 11 \log_2 \lambda \rceil}, \qquad L = \kappa = \lambda.$$

Instead of the earlier illustrative choice $t = \lambda^4$, set t to the exact minimum required to make the residual-hashing upper bound at most $2^{-\lambda}$:

$$\boxed{t = ((h+1)L + 1)p + 2(\lambda - 1).}$$

Then

$$t = \Theta(\lambda^3 \log \lambda), \qquad M = tL = \Theta(\lambda^4 \log \lambda).$$

The compact public-key payload is therefore

$$\boxed{|pk_C| = \kappa_A + tL(p - q) = \Theta(\lambda^4 (\log \lambda)^2)}$$

for any κ_A below that order. This improves the earlier illustrative compact-key growth $\Theta(\lambda^5 \log \lambda)$ by a factor $\Theta(\lambda / \log \lambda)$.

The source width α_0 above does *not* by itself define the integer error law that reaches the WC layer after binary-secret and modulus transformations. We therefore make the final-noise obligation explicit rather than silently deriving it. Let χ_λ denote the integer error law delivered to each of the $M = tL$ WC coordinates after all external source transformations. Assume that these errors satisfy the joint independence conditions of Theorem 9.1 and, for some absolute constant C_B,

$$B_\lambda \le C_B \lambda^5 (\log \lambda)^{5/2}, \qquad \delta_E(\lambda) := \Pr_{E \leftarrow \chi_\lambda}(|E| \le B_\lambda) \ge 1 - \lambda^{-6}.$$

These are additional premises on the *final* integer error, not consequences asserted from α_0 in this paper. Under them,

$$\frac{MB_\lambda}{Q} = O\Big(\lambda^{-2} (\log \lambda)^{7/2}\Big) = o(1), \qquad M(1 - \delta_E) = O(\lambda^{-2} \log \lambda) = o(1),$$

so $2B_\lambda < Q$ eventually and $\overline{D}_R = 1 + o(1)$. Perfect correctness has margin

$$\frac{tQ}{P} = O(\lambda^{-2} \log \lambda) = o(1),$$

and the residual term is exactly bounded by $2^{-\lambda}$ by construction.

Corollary 12.1 (Conditional asymptotic compatibility)**.** *Assume (i) a binary-secret LWE source satisfying the stated resource bounds is hard for the family above, (ii) the external transformations produce final independent symmetric integer errors χ_λ satisfying the explicit $B_\lambda, \delta_E(\lambda)$ bounds above across all $M = tL$ coordinates, and (iii) all adversarial oracle/query budgets are polynomial. Then the terms in Theorem 11.1 are negligible, and MODDIV-WC-FAST-COMPACT-FO has a conditional IND-CCA route in the single-user classical programmable random-oracle model. This corollary does not prove premise (ii) from the source width α_0, nor does it supply a finite concrete-security estimate.*

This is an asymptotic compatibility statement for the displayed inequalities. Small numerical values of λ need not satisfy every external source precondition; no statement of "128-bit security" follows by substituting $\lambda = 128$.

13 Parameter Diet and a Proof-Route Size Barrier

This section asks whether parameter optimization alone can make the compact scheme practical. The answer for the present proof inequalities is negative.

Write $g = p - q$ and suppose the final CCA upper bound is required to be at most $2^{-\tau}$. Even before accounting for the actual LWE advantage, a necessary condition from

$$q_H 2^{-\kappa} + N\sqrt{\overline{D}_R\,(a + 2^{-\kappa} + \varepsilon_{\mathrm{enc}})} \le 2^{-\tau}$$

is, after dropping the outer fallback term $q_H 2^{-\kappa}$, setting $N = 1$, $\overline{D}_R = 1$, $a = 0$, and dropping programming losses while retaining the inner message-guessing term $2^{-\kappa}$,

$$\kappa \ge 2\tau.$$

Since $L \ge \kappa$, the most favorable relaxed choice has $L = 2\tau$. The residual-hashing bound additionally forces, even after setting the structural dimension to the unrealistically favorable $h = 1$,

$$t \ge ((h+1)L + 1)p + 4\tau - 2.$$

Correction supplies an upper bound on t, and their interaction yields the following proof-route lower bound.

Theorem 13.1 (Relaxed lower bound certified by the present inequalities)**.** *For integer $\tau \ge 1$, any parameter tuple for which the displayed CCA upper bound can certify an advantage at most $2^{-\tau}$ has public payload at least*

$$K_{\min}^{\mathrm{relax}}(\tau) = 2\tau\, g_*\Big((4\tau+1)(g_*+1) + 4\tau - 2\Big)$$

bits, where g_* is the smallest integer $g \ge 3$ satisfying

$$(4\tau+1)(g+1) + 4\tau - 2 \le 2^{g-2} - 1.$$

The bound is obtained after relaxing to $h = 1$, zero LWE advantage, $\overline{D}_R = 1$, $N = 1$, dropping the outer fallback-guessing term and programming losses, and allowing zero seed cost; the $2^{-\kappa}$ term inside the OW square root is retained; it is therefore a necessary lower bound for what the current proof inequalities can certify, not a lower bound on every possible security proof for MODDIV-WC.

Proof. Under the stated relaxation, $\kappa, L \ge 2\tau$. For fixed $g = p - q$, the smallest possible p obeys $p \ge g + 1$ because $q \ge 1$. Hence the residual constraint gives

$$t \ge T_0(g) := (4\tau+1)(g+1) + 4\tau - 2.$$

Perfect correctness implies, for all $q \ge 1$,

$$t < \frac{2^{g-3}}{1 - 2^{-q}} \le 2^{g-2},$$

so integer feasibility requires $T_0(g) \le 2^{g-2} - 1$ and hence $g \ge g_*$. The public payload tLg is therefore at least $T_0(g_*) \cdot 2\tau \cdot g_*$; the product is increasing for $g \ge 3$. □

For $\tau = 128$, the relaxed lower bound is about 4.51 MiB. This already excludes a roughly kilobyte-scale key *as certified by the current proof formula*, before the true $h = \Omega(n_0^2)$ embedding cost, Renyi losses, query budgets, seed, or a nonzero LWE advantage are restored. It does not show that the cryptosystem is attacked at smaller sizes, nor that a future tighter proof could not do substantially better.

Table 1: Relaxed public-payload lower bounds imposed by the present proof inequalities.

τ	g_*	$T_0(g_*)$	minimum bits	minimum bytes
32	13	1,932	1,607,424	200,928
64	15	4,366	8,382,720	1,047,840
128	16	9,231	37,810,176	4,726,272
192	16	13,839	85,026,816	10,628,352
256	17	19,472	169,484,288	21,185,536

14 Revision Relative to the Public ePrint Version

The first public MODDIV-WC manuscript is ePrint 2026/2747 [3]. The present A4 v1.2 revision changes the public technical record in eight specific ways:

- the perfect-correctness proof now uses the signed congruence $v_i \equiv w_i + \delta_i \pmod{P}$ and $|\delta_i| < P/8$, rather than an invalid ordinary difference of canonical representatives;
- FAST is stated as an exact identity and is separated from all hardness assumptions;
- COMPACT is formulated as an oracle-coupled representation theorem, not as a standard-model PRG claim for a public seed;
- the implicit-rejection composition counts the plaintext-checking and oracle calls induced by decapsulation required by the corrected HHK interface;
- the numerical parameter section distinguishes an asymptotic compatibility result from a finite concrete-security certificate and proves a lower bound only for what the written proof formula can certify;
- the Renyi theorem now states independence of the final error from the public ribbons and gives the exact dangerous-residue set;
- KEM key generation now samples one persistent fallback secret per key, matching the implicit-rejection interface;
- the asymptotic corollary now states explicit quantitative premises for the final integer error law, and the reproducibility package includes a strict reference wire parser and functional KEM rejection tests.

These changes do not turn the historical ModDiv constructions into instances of the WC security theorem.

15 Reproducibility Artifact

The public companion archive contains only this article, public-source notes, and self-contained verification code. The primary checker validates the unit-anchor fibers, sparse embedding, parity-mask law, FAST identity, reconciliation, the signed-wraparound correctness witness, honest BASE transcripts on toy profiles, the independent-noise Renyi witness, the strict canonical wire format, persistent fallback behavior, functional KEM re-encryption, payload arithmetic, exact threshold formulas, and the proof-route barrier table. A second checker re-derives selected claims without importing the primary arithmetic module.

Run

```
python3 verify_a4.py --output A4_RESULTS.json
python3 verify_a4_independent.py --output A4_INDEPENDENT.json
```

The deterministic SHAKE256 expansion in the code is a functional test instance of COMPACT only; it is not a proof that SHAKE256 is a programmable random oracle. Finite tests do not establish LWE hardness or adaptive CCA security.

16 Limitations and Open Problems

The result is deliberately proof-oriented. The following limitations are explicit.

- **Concrete practicality.** The current proof route does not yield competitive public-key sizes. Theorem 13.1 shows that parameter tuning alone cannot reach kilobyte-scale certification for a 2^{-128} target under the present inequalities.
- **Proof tightness.** The dominant structural costs are the residual-hashing requirement on t and the binary-secret/modulus-reduction route that makes h large. A substantially tighter OW-CPA proof or a different embedding could change the size barrier.
- **Concrete LWE estimation.** No parameter set is labelled "128-bit secure". The finite arithmetic examples are checks of the written inequalities, not attack-cost estimates.
- **Concrete XOF.** COMPACT uses a programmable classical random oracle. SHAKE256 is only an engineering instantiation and does not realize that programming argument in the standard model.
- **QROM and multi-user security.** Neither is proved here.
- **Implementation security.** The package now supplies a strict reference parser/serializer and functional KEM rejection tests, but it remains research code. Constant-time behavior, side channels, fault attacks, denial-of-service bounds, and production hardening remain outside the theorem.
- **Independent review.** The construction, the masked-Hankel Renyi lemma, the compact coupling, and the complete theorem chain require external cryptanalysis and peer review.

The most consequential open theoretical question is no longer simple parameter tuning. It is whether one can replace the residual-hashing/embedding modules by a tighter argument while retaining the exact unit-anchor and rounding structure. Until such a proof is found, MODDIV-WC should be viewed as a mathematically grounded prototype and proof vehicle rather than a practical KEM candidate.

17 Conclusion

MODDIV-WC Version 2 separates four questions that were intertwined in the first public version: correctness, arithmetic evaluation, public-key representation, and security composition. The unit anchor $X_0 = 1$ still provides every-key surjectivity and supports the exact parity-mask binary-LWE embedding. The new FAST theorem shows that the rounded Hankel output can be evaluated by a high-part sum plus the exact low-part carry, with no cryptographic approximation. The new COMPACT representation replaces the explicit random Hankel inputs by a short public seed in a dedicated programmable classical random oracle, reducing the transmitted public key to $\kappa_A + tL(p-q)$ bits. Replaying HHK with independent $\mathcal{A}, G, H$ domains, a fallback secret sampled once at KEM key generation, and strict canonical bytes preserves the conditional CCA route in the principal single-user classical-ROM game without additional FAST loss and with only the stated COMPACT prequery coupling when applicable.

The parameter-diet analysis also supplies the main negative result of Version 2. The earlier illustrative choice of t was unnecessarily large and can be reduced asymptotically, but the

present proof remains intrinsically expensive. The residual-hashing square-root loss forces large κ, L, t, while the cited binary-secret route makes h quadratic in the source dimension. Even after relaxing away the actual lattice cost, Renyi loss, query budgets, and seed size, the current inequalities cannot certify a 2^{-128} target with a kilobyte-scale public key. This should be read as a diagnosis of the proof architecture, not as an attack or a universal lower bound on ModDiv constructions.

Accordingly, Version 2 presents MODDIV-WC as a coherent proof-oriented construction with separately specified BASE, FAST, COMPACT, and conditional classical-ROM CCA layers. The parameter analysis explains why the present proof route is not practically competitive. Further progress would require a materially tighter proof module, a stronger direct source theorem, or a different architecture rather than another round of parameter trimming.

AI Contribution Statement

This work was developed with substantial assistance from ChatGPT (GPT-5.6 Sol, OpenAI), under the scientific direction of Samir Bouftass. ChatGPT contributed materially to the mathematical development and adversarial analysis of MODDIV-WC, including analysis of the binary-LWE embedding, Renyi–Hankel transfer, exact FAST identity, programmable-ROM compact representation, HHK game composition, proof-route inequalities, reproducibility code, and manuscript preparation. Samir Bouftass defined and directed the ModDiv research program, supplied the underlying constructions and prior work, selected the research objectives, challenged intermediate claims, and supervised the iterative review. The Version-2 claims remain research claims and require independent cryptanalysis and review before cryptographic reliance or deployment.

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